

线性代数中的梯度计算

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1 从定义出发

$$1.1 \quad e = \mathbf{x}^T A \mathbf{x}, A^T = A, \dim(A) = (N, N), \dim(\mathbf{x}) = (N, 1)$$

$$\begin{aligned} e + de &= (\mathbf{x} + d\mathbf{x})^T A (\mathbf{x} + d\mathbf{x}) \\ de &= d\mathbf{x}^T A \mathbf{x} + \mathbf{x}^T A d\mathbf{x} + d\mathbf{x}^T A d\mathbf{x} \\ &= d\mathbf{x}^T A \mathbf{x} + \mathbf{x}^T A d\mathbf{x} + O(d\mathbf{x}) \\ &= \mathbf{x}^T A d\mathbf{x} + \mathbf{x}^T A d\mathbf{x} \\ &= 2\mathbf{x}^T A d\mathbf{x} \\ \frac{de}{d\mathbf{x}} &= 2\mathbf{x}^T A \end{aligned}$$

$$1.2 \quad e = (\mathbf{y} - A\mathbf{x})^T (\mathbf{y} - A\mathbf{x}), \dim(A) = (M, N), \dim(\mathbf{x}) = (N, 1)$$

$$\begin{aligned} e + de &= (\mathbf{y} - A(\mathbf{x} + d\mathbf{x}))^T (\mathbf{y} - A(\mathbf{x} + d\mathbf{x})) \\ &= (\mathbf{y} - A\mathbf{x} - A d\mathbf{x})^T (\mathbf{y} - A\mathbf{x} - A d\mathbf{x}) \\ de &= (-A d\mathbf{x})^T (\mathbf{y} - A\mathbf{x}) + (\mathbf{y} - A\mathbf{x})^T (-A d\mathbf{x}) + (A d\mathbf{x})^T (A d\mathbf{x}) \\ &= (-A d\mathbf{x})^T (\mathbf{y} - A\mathbf{x}) + (\mathbf{y} - A\mathbf{x})^T (-A d\mathbf{x}) + O(d\mathbf{x}) \\ &= (\mathbf{y} - A\mathbf{x})^T (-A d\mathbf{x}) + (\mathbf{y} - A\mathbf{x})^T (-A d\mathbf{x}) \\ &= 2(A\mathbf{x} - \mathbf{y})^T A d\mathbf{x} \\ \frac{de}{d\mathbf{x}} &= 2(A\mathbf{x} - \mathbf{y})^T A \end{aligned}$$

2 各分量偏导数

2.1 $e = \mathbf{x}^T A \mathbf{x}, A^T = A, \dim(A) = (N, N) \dim(\mathbf{x}) = (N, 1)$

$$\begin{aligned}
 e &= \sum_{i=1}^N \sum_{j=1}^N x_i a_{i,j} x_j \\
 \frac{\partial e}{\partial x_k} &= \sum_{i=1}^N \sum_{j=1}^N \frac{\partial (x_i a_{i,j} x_j)}{\partial x_k} \\
 &= \sum_{i=1}^N \sum_{j=1}^N \left(\frac{\partial x_i}{\partial x_k} a_{i,j} x_j + x_i a_{i,j} \frac{\partial x_j}{\partial x_k} \right) \\
 &= \sum_{i=1}^N \sum_{j=1}^N \frac{\partial x_i}{\partial x_k} a_{i,j} x_j + \sum_{i=1}^N \sum_{j=1}^N x_i a_{i,j} \frac{\partial x_j}{\partial x_k} \\
 &= \sum_{j=1}^N a_{k,j} x_j + \sum_{i=1}^N x_i a_{i,k} \\
 &= \sum_{i=1}^N a_{k,i} x_i + \sum_{i=1}^N x_i a_{i,k} \\
 &= \sum_{i=1}^N (a_{k,i} + a_{i,k}) x_i \\
 &= 2 \sum_{i=1}^N x_i a_{i,k} \\
 &= 2 \mathbf{x}^T A_{[:,k]} \quad A_{[:,k]} = (a_{1,k} \ a_{2,k} \ \cdots \ a_{N,k})^T
 \end{aligned}$$

2.2 $e = (\mathbf{y} - A \mathbf{x})^T (\mathbf{y} - A \mathbf{x}), \dim(A) = (M, N) \dim(\mathbf{x}) = (N, 1)$

$$\begin{aligned}
 e &= \sum_{i=1}^M (y_i - \sum_{j=1}^N a_{i,j} x_j)^2 \\
 \frac{\partial e}{\partial x_k} &= \sum_{i=1}^M \frac{\partial (y_i - \sum_{j=1}^N a_{i,j} x_j)^2}{\partial x_k} \\
 &= 2 \sum_{i=1}^M (y_i - \sum_{j=1}^N a_{i,j} x_j) \frac{\partial (y_i - \sum_{j=1}^N a_{i,j} x_j)}{\partial x_k}
 \end{aligned}$$

$$\begin{aligned}
&= 2 \sum_{i=1}^M (y_i - \sum_{j=1}^N a_{i,j} x_j) \frac{\partial (-\sum_{j=1}^N a_{i,j} x_j)}{\partial x_k} \\
&= 2 \sum_{i=1}^M (\sum_{j=1}^N a_{i,j} x_j - y_i) \sum_{j=1}^N \frac{\partial (a_{i,j} x_j)}{\partial x_k} \\
&= 2 \sum_{i=1}^M (\sum_{j=1}^N a_{i,j} x_j - y_i) \sum_{j=1}^N a_{i,j} \frac{\partial x_j}{\partial x_k} \\
&= 2 \sum_{i=1}^M (\sum_{j=1}^N a_{i,j} x_j - y_i) a_{i,k} \\
&= 2 \sum_{i=1}^M (A_{[i,:]} \mathbf{x} - y_i) a_{i,k} \quad A_{[i,:]} = (a_{i,1} \ a_{i,2} \ \cdots \ a_{i,N}) \\
&= 2(A\mathbf{x} - \mathbf{y})^T A_{[:,k]} \quad A_{[:,k]} = (a_{1,k} \ a_{2,k} \ \cdots \ a_{M,k})^T
\end{aligned}$$

3 向量求导

3.1 $\mathbf{y} = A\mathbf{x}$

$$\frac{d\mathbf{y}}{d\mathbf{x}} = A$$

3.2 $Y = AU + BV, \dim(U) = (N, 1), \dim(V) = (M, 1)$

$$\begin{aligned}
\frac{\partial Y}{\partial U} &= A \\
\frac{\partial Y}{\partial V} &= B
\end{aligned}$$

3.3 $\mathbf{y} = A\mathbf{u} + B\mathbf{v}, \mathbf{u} = C\mathbf{x}, \mathbf{v} = D\mathbf{x}, \dim(\mathbf{x}) = (N, 1)$

$$\begin{aligned}
\frac{d\mathbf{u}}{d\mathbf{x}} &= \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{x}} + \frac{\partial \mathbf{y}}{\partial \mathbf{v}} \frac{d\mathbf{v}}{d\mathbf{x}} \\
&= AC + BD
\end{aligned}$$

$$3.4 \quad e = \mathbf{x}^T A \mathbf{x}, A^T = A, \dim(\mathbf{x}) = (N, 1)$$

设

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= \mathbf{u}^T A \mathbf{v} \\ \mathbf{u} &= \mathbf{x} \\ \mathbf{v} &= \mathbf{x} \end{aligned}$$

得

$$\begin{aligned} \frac{d\mathbf{y}}{d\mathbf{x}} &= \frac{\partial(\mathbf{u}^T A \mathbf{v})}{\partial \mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{x}} + \frac{\partial(\mathbf{u}^T A \mathbf{v})}{\partial \mathbf{v}} \frac{d\mathbf{v}}{d\mathbf{x}} \\ &= \frac{\partial(\mathbf{v}^T A^T \mathbf{u})}{\partial \mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{x}} + \mathbf{u}^T A \frac{d\mathbf{v}}{d\mathbf{x}} \\ &= \frac{\partial(\mathbf{v}^T A \mathbf{u})}{\partial \mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{x}} + \mathbf{u}^T A \frac{d\mathbf{v}}{d\mathbf{x}} \\ &= \mathbf{v}^T A + \mathbf{u}^T A \\ &= \mathbf{x}^T A + \mathbf{x}^T A \\ &= 2\mathbf{x}^T A \end{aligned}$$

$$3.5 \quad e = (\mathbf{y} - A\mathbf{x})^T (\mathbf{y} - A\mathbf{x}), \dim(A) = (M, N), \dim(\mathbf{x}) = (N, 1)$$

$$\begin{aligned} e &= \mathbf{u}^T \mathbf{u} & \mathbf{u} &= \mathbf{y} - A\mathbf{x} \\ \frac{de}{d\mathbf{x}} &= \frac{de}{d\mathbf{u}} \frac{d\mathbf{u}}{d\mathbf{x}} \\ &= (2\mathbf{u})^T (-A) \\ &= 2(\mathbf{y} - A\mathbf{x})^T (-A) \\ &= 2(A\mathbf{x} - \mathbf{y})^T A \end{aligned}$$